Gradients, Tangents & Normals

Exam code: 9MA0

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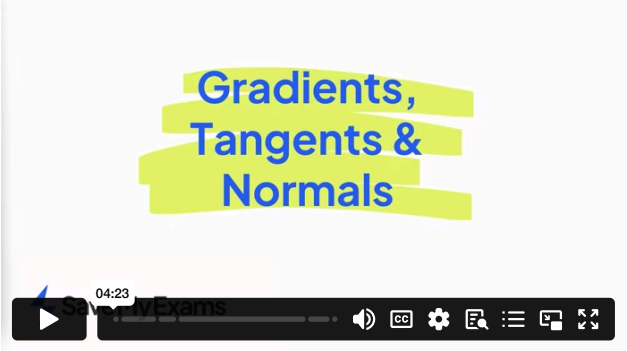
Updated on 20 June 2025



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Help me understand

Gradients, tangents & normals

How do I find the gradient at a point on a curve?

- To find the gradient of a curve **y= f(x)** at any point on the curve, substitute the ***x*-coordinate** of the point into the derivative **f'(x)**

e.g. FIND THE GRADIENT OF THE CURVE $y = x - \frac{1}{x}$ AT THE POINT $(2, \frac{3}{2})$

$y = x - x^{-1}$

$\frac{dy}{dx} = 1-(-1)x^{-1-1} = 1+x^{-2} = 1 + \frac{1}{x^2}$

FIND DERIVATIVE

WHEN $x = 2$,

$\frac{dy}{dx} = 1 + \frac{1}{2^2} = \frac{5}{4}$

PUT x–COORDINATE INTO DERIVATIVE

THE GRADIENT AT $(2, \frac{3}{2})$ IS $\frac{5}{4}$

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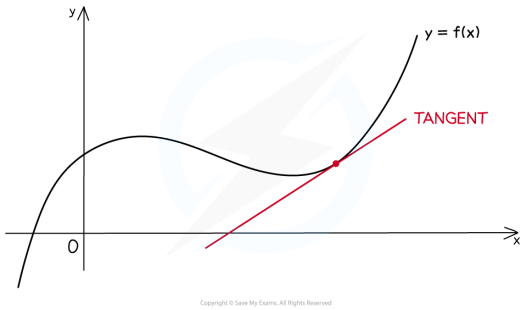
How do I find the equation of a tangent to a curve?

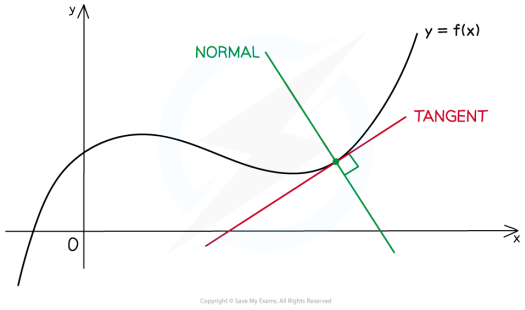
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



- For the curve y = f(x), you can find the equation of the **tangent** at the point (a, f(a)) using $y - \mathrm{f}(a) = \mathrm{f}'(a)(x - a)$

How do I find the equation of a normal to a curve?

- At any point on a curve, the **normal** is the line that goes through the point and is **perpendicular** to the tangent at that point



- For the curve y = f(x), you can find the equation of the **normal** at the point (a, f(a)) using $y - \mathrm{f}(a) = -\frac{1}{\mathrm{f}'(a)}(x - a)$

Examiner Tips and Tricks

- The formulae above are not in the exam formulae booklet, but if you understand what tangents and normals are, then the formulae follow from the equation of a straight line combined with parallel and perpendicular gradients (see Worked Example below).

Worked Example

a) Find the equation of the tangent to C at the point where $x = 9$.

b) Find the equation of the normal to C at the same point.

a) WHEN $x = 9$, $y = 7 - 12\sqrt{9} = -29$ ← FIND y–COORDINATE

$y = 7 - 12x^{\frac{1}{2}}$ ← $\sqrt{x} = x^{\frac{1}{2}}$

$\frac{dy}{dx} = -12\left(\frac{1}{2}\right)x^{\frac{1}{2}-1} = -6x^{-\frac{1}{2}} = -\frac{6}{\sqrt{x}}$ ← FIND DERIVATIVE

WHEN $x = 9$,

$\frac{dy}{dx} = -\frac{6}{\sqrt{9}} = -2$ ← USE DERIVATIVE TO FIND GRADIENT OF TANGENT

SO THE TANGENT AT POINT $(9, -29)$ HAS EQUATION

$y - (-29) = -2(x - 9)$ ← $y - y_1 = m(x - x_1)$

$y = -2x - 11$

b) THE NORMAL IS PERPENDICULAR TO THE TANGENT, SO THE GRADIENT OF THE NORMAL IS $-\frac{1}{(-2)} = \frac{1}{2}$

AT $(9, -29)$ THE NORMAL HAS EQUATION

$m_2 = -\frac{1}{m_1}$ FOR PERPENDICULAR LINES

$y - (-29) = \frac{1}{2}(x - 9)$

$y = \frac{1}{2}x - \frac{67}{2}$

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